

Resources

REAL VARIABLES I

Fall 2026

For every topic we cover, I will provide either lecture notes or a precise reading reference. The books below offer complementary treatments; you do not need to purchase them.

0.1 Main textbook

- Luigi Ambrosio, Giuseppe Da Prato, and Andrea Menzucchi, *Introduction to Measure Theory and Integration*, Publications of the Scuola Normale Superiore 10, Edizioni della Normale, 2011. This is our principal reference. [Publisher page and electronic access](#).

0.2 Additional references

- Elias M. Stein and Rami Shakarchi, *Real Analysis: Measure Theory, Integration, and Hilbert Spaces*, Princeton Lectures in Analysis III, Princeton University Press, 2005. [Book record](#).
- Sheldon Axler, *Measure, Integration & Real Analysis*, Graduate Texts in Mathematics 282, Springer, 2020. The author provides a legal, free electronic edition. [Open-access book](#).
- Gerald B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999. This is a more advanced reference with extensive coverage beyond the course. [Publisher page](#).

0.3 Freely available notes and texts

The following materials are available online and are also represented in the course **books/** archive. They are optional supplementary reading, not required purchases.

- Jeff Vialovskiy, *18.125 Measure and Integration* (MIT OpenCourseWare, Fall 2003). A 24-part set of lecture notes. [Online lecture notes](#).
- Brian S. Thomson, Judith B. Bruckner, and Andrew M. Bruckner, *Real Analysis*, 2nd ed., 2008. [Free PDF](#).
- Steve Cheng, *A Crash Course on the Lebesgue Integral and Measure Theory*. [Online PDF](#).
- D. H. Fremlin, *Measure Theory*, Volumes 1–5. The first two volumes are especially suitable as introductory references. [Online contents and PDFs](#).
- Terence Tao, *An Introduction to Measure Theory*. [Author's preliminary PDF](#).
- John K. Hunter, *Measure Theory*. [Online PDF](#).
- John E. Hutchinson, *Analysis 2, 2012: Lecture Notes*. [Online PDF](#).
- Jeffrey Steif, *Lecture Notes on Measure Theory, Integration Theory and Differentiation Theory*. [Online PDF](#).

0.4 Office hours

Office hours are part of the course. Bring questions about definitions, proofs, examples, reading, or homework - even when your question is only "where should I start?"

Discussing homework in office hours is encouraged. If you give a clear, complete solution of a designated homework exercise, I may propose to you a grade for that exercise during the meeting. If you are satisfied with the grade, you will not need to submit a written solution for that exercise.

0.5 Access and well-being

- [Center for Educational Access](#)
- [Counseling and Psychological Services](#)
- [U of A Cares](#)
- [University emergency information](#)
- [RazALERT](#)